



SIL Verification Report

Hybrit Pilot Plant

REV	DATE	APPROVED	DESCRIPTION OF CHANGE
	5/26/2020		Document created (First calculation of SIF)
1	6/25/2020		Change from E&H Ex Barrier to PR electronics.

Automatically generated by exSILentia® version 4.8.4.0.

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1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the results, selections, and assumptions made during the Safety Integrity Level verification for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project.

1.1 General Project Information

Project Identification: 001
Project Name: Hybrit Pilot Plant
Revision: 0
Company:
Project Description:

2 SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högspänningsenhet (345EC015)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högspänningsenhet (345EC015) Safety Instrumented Function.

2.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningsenhet (345EC015)
SIF Tag	SIF-058
SIF Description:	SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningsenhet (345EC015)

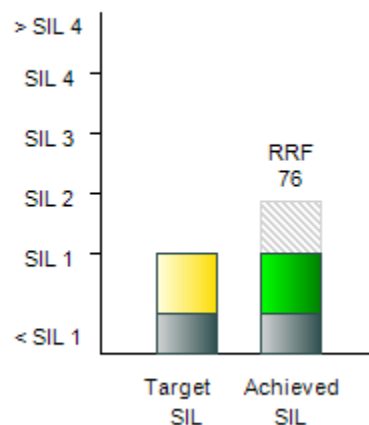
2.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 76.7



2.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högspänningsenhet (345EC015) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

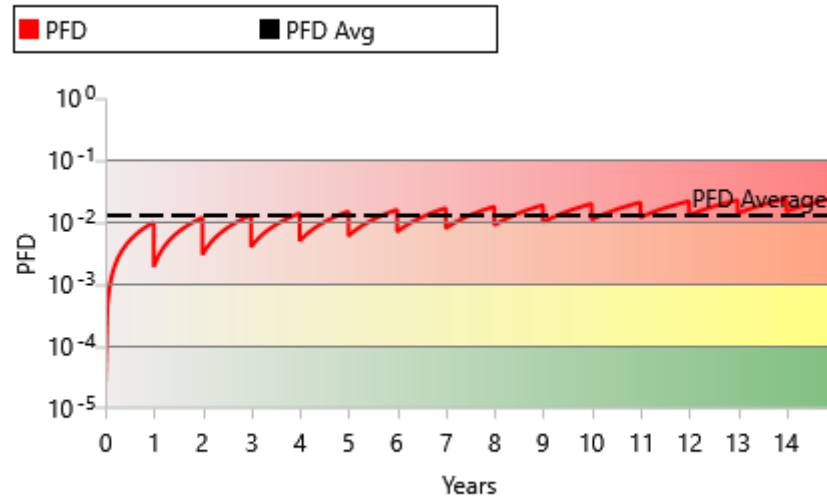
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högsäpänningsenhet (345EC015)



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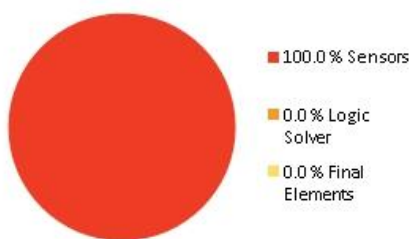
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högsäpänningsenhet (345EC015) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

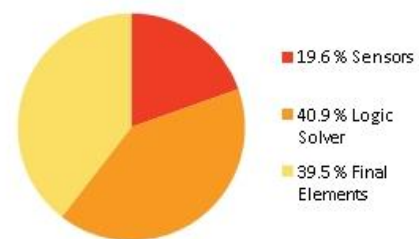
Table 1 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.30E-02	76.7	1	2	N/A	29.66

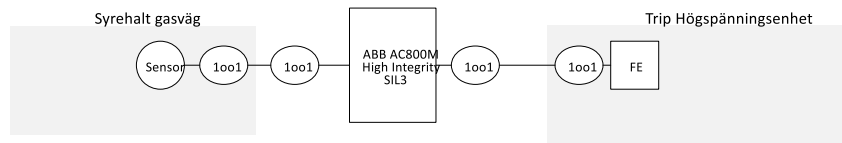
PFDavg Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



2.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högsäpänningsenhet (345EC015) Safety Instrumented Function is quantified as follows.

Table 2 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.30E-02	2	N/A	151.42	0	2

Number of Sensor group(s): 1
 Voting between groups: 1001
 β-factor [%]: N/A

2.3.1.1 Sensor Group 1: Syrehalt gasväg

The information and reliability data underneath describe the Syrehalt gasväg sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
 HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Siemens Sitrans SL		-	-	B	1

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 72

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Syrehalt gasväg.

Table 3 Reliability Data Sensor Group Syrehalt gasväg

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Siemens Sitrans SL	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.41E-07 <i>7.41E-07</i>	0.00E+00 <i>0.00E+00</i>	1.16E-06 <i>1.16E-06</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		39.6
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

2.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högsäpningssenhett (345EC015) Safety Instrumented Function is quantified as follows.

Table 4 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.10E-08	3	N/A	72.47	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:	Architecture Type:	B					
HFT:	β-factor [%]:	1					
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 5 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

2.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-058 SIL1 Syrehalt gasväg (345AZ362|H2) => Trip Högsäpänningsenhet (345EC015) Safety Instrumented Function is quantified as follows.

Table 6 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.15E-06	2	N/A	75.1	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

2.3.3.1 Final Element Group 1: Trip Högsäpänningsenhet

The information and reliability data underneath describe the Trip Högsäpänningsenhet final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
PSR-PS20		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip Högsäpänningsenhet.

Table 7 Reliability Data Final Element Group Trip Högspänningsenhet

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
PSR-PS20		1.42E-06		1.95E-10			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

3 SIF-059 SIL2 Vätsketryck nedströms pump ($345PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(345SV566)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-059 SIL2 Vätsketryck nedströms pump ($345PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(345SV566) Safety Instrumented Function.

3.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Vätsketryck nedströms pump ($345PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(345SV566)
SIF Tag	SIF-059
SIF Description:	SIL2 Vätsketryck nedströms pump ($345PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(345SV566)

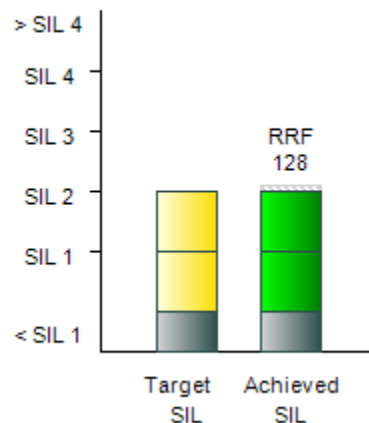
3.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 128.9



3.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-059 SIL2 Vätsketryck nedströms pump ($345PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(345SV566) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

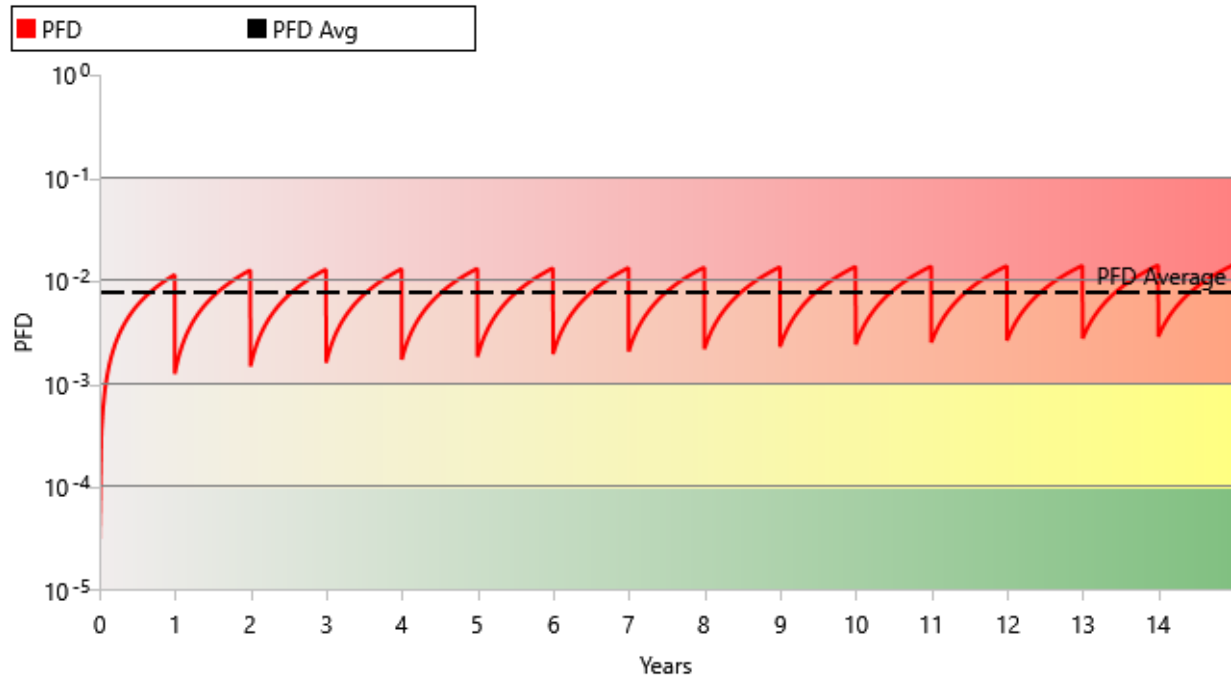
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566)



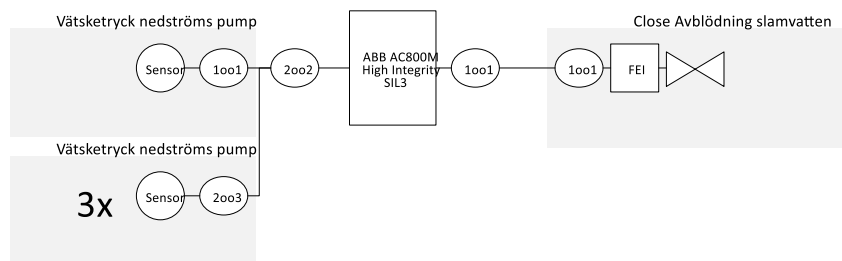
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-059 SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 8 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.75E-03	128.9	2	2	N/A	64.98



The SIF Conceptual Design is modeled as shown in the picture below.



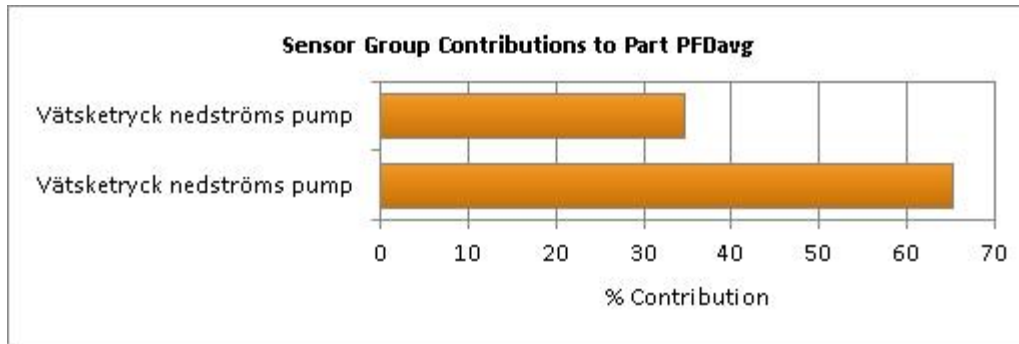
3.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-059 SIL2 Vätsketryck nedströms pump ($345PI326 < (320PZ320/1/2 + 0,5 \text{ bar}) \Rightarrow \text{Close Avblödning slamvatten}(345SV566)$) Safety Instrumented Function is quantified as follows.

Table 9 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.20E-03	2	N/A	10362.29	0	2

Number of Sensor group(s): 2
Voting between groups: 2oo2
β-factor [%]:



3.3.1.1 Sensor Group 1: Vätsketryck nedströms pump

The information and reliability data underneath describe the Vätsketryck nedströms pump sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor: N/A
[%]:

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätsketryck nedströms pump.

Table 10 Reliability Data Sensor Group Vätsketryck nedströms pump

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

3.3.1.2 Sensor Group 2: Vätsketryck nedströms pump

The information and reliability data underneath describe the Vätsketryck nedströms pump sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): „

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätsketryck nedströms pump.

Table 11 Reliability Data Sensor Group Vätsketryck nedströms pump

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	18.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

3.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-059 SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566) Safety Instrumented Function is quantified as follows.

Table 12 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.84E-08	3	N/A	72.03	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 13 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

3.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-059 SIL2 Vätstektryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566) Safety Instrumented Function is quantified as follows.

Table 14 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

3.3.3.1 Final Element Group 1: Close Avblödning slamvatten

The information and reliability data underneath describe the Close Avblödning slamvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worchester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avblödning slamvatten.

Table 15 Reliability Data Final Element Group Close Avblödning slamvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worchester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	12.6
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

4 SIF-060 SIL2 Vätskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-060 SIL2 Vätskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function.

4.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tillopp spolvatten (345SV634)
SIF Tag	SIF-060
SIF Description:	SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tillopp spolvatten (345SV634)

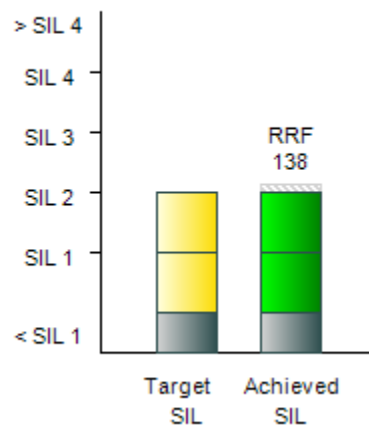
4.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 138.4



4.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-060 SIL2 Vätskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

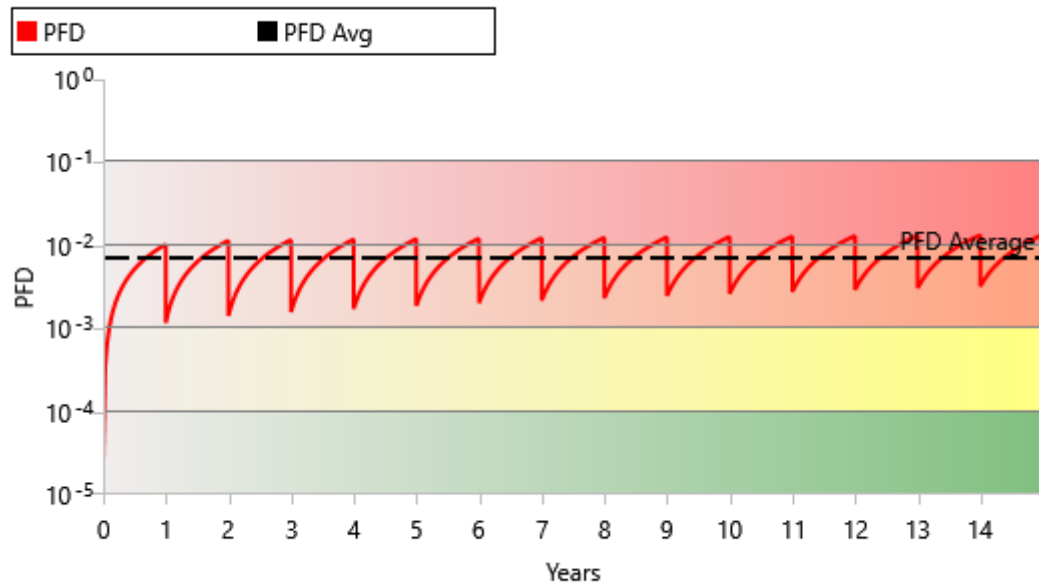
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

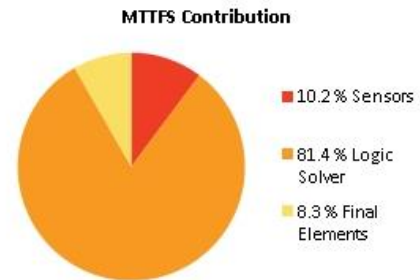
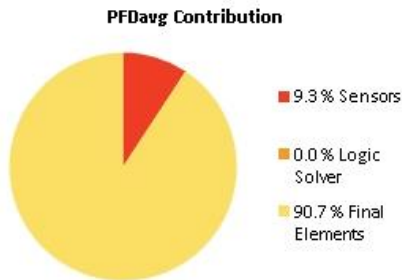
SIL2 Vättskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634)



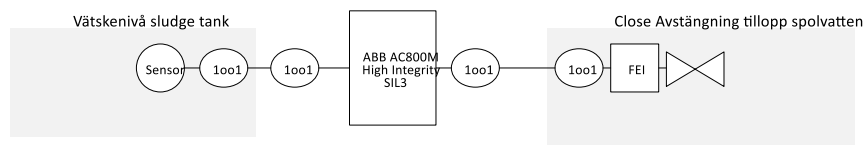
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-060 SIL2 Vättskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 16 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.22E-03	138.4	2	2	N/A	59.02



The SIF Conceptual Design is modeled as shown in the picture below.



4.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-060 SIL2 Vätskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function is quantified as follows.

Table 17 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.69E-04	2	N/A	576.96	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

4.3.1.1 Sensor Group 1: Vätskenivå sludge tank

The information and reliability data underneath describe the Vätskenivå sludge tank sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

Endress+Hauser Liquiphant M/S + FEL 58 2006.2.02 - - B -

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vättskenivå sludge tank.

Table 18 Reliability Data Sensor Group Vättskenivå sludge tank

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	8.39E-08 <i>9.01E-08</i>	8.53E-08 <i>8.53E-08</i>	6.21E-09 <i>0.00E+00</i>	5.92E-08 <i>5.92E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	77.2
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

4.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-060 SIL2 Vättskenivå sludge tank (345LS310|H2) => Close Avstängning tilllopp spolvatten (345SV634) Safety Instrumented Function is quantified as follows.

Table 19 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.10E-08	3	N/A	72.47	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β -factor [%]: 1
MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 20 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

4.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-060 SIL2 Vättskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function is quantified as follows.

Table 21 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

4.3.3.1 Final Element Group 1: Close Avstängning tillopp spolvatten

The information and reliability data underneath describe the Close Avstängning tillopp spolvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängning tillopp spolvatten.

Table 22 Reliability Data Final Element Group Close Avstängning tillopp spolvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	12.6
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

5 SIF-061 SIL2 Vätskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-061 SIL2 Vätskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566) Safety Instrumented Function.

5.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566)
SIF Tag	SIF-061
SIF Description:	SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566)

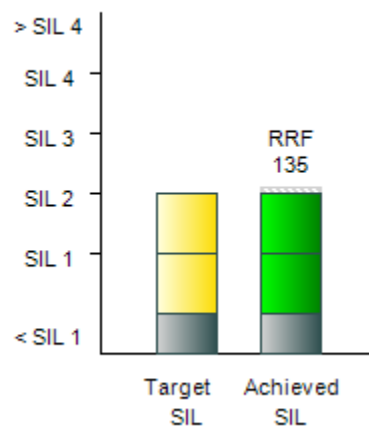
5.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 135.5



5.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-061 SIL2 Vätskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

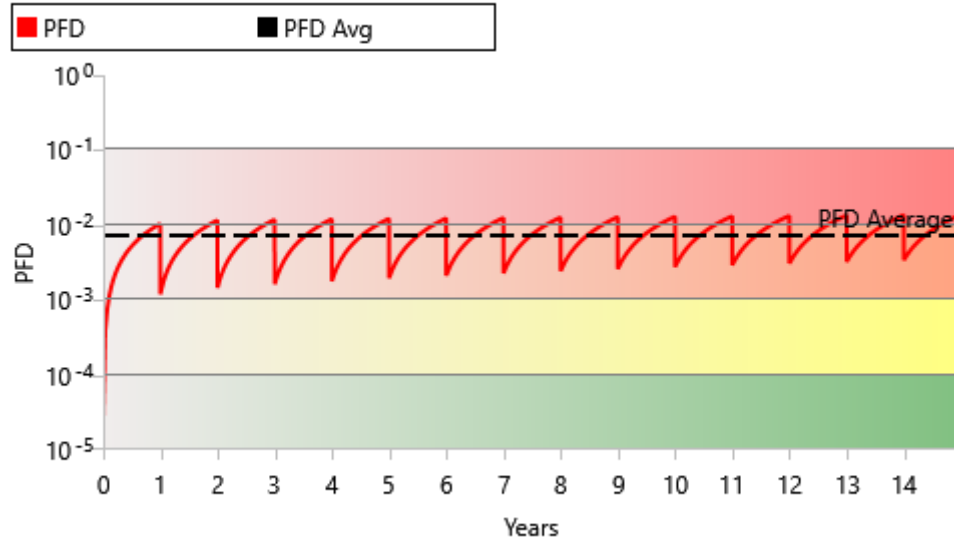
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Vätskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566)



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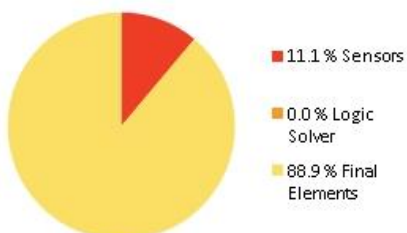
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-061 SIL2 Vätskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

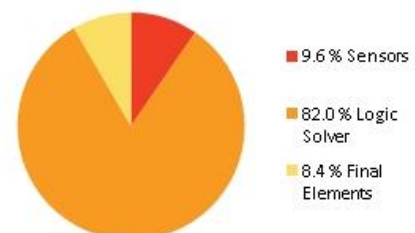
Table 23 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.38E-03	135.5	2	2	N/A	59.46

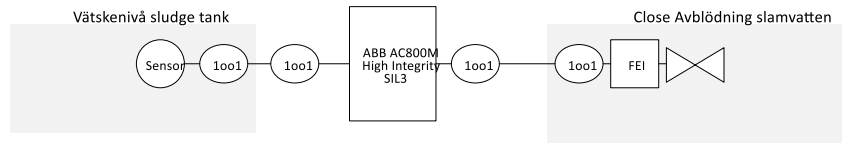
PFDavg Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



5.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-061 SIL2 Vätskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566) Safety Instrumented Function is quantified as follows.

Table 24 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.22E-04	2	N/A	621.33	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

5.3.1.1 Sensor Group 1: Vätskenivå sludge tank

The information and reliability data underneath describe the Vätskenivå sludge tank sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Liquiphant M/S + FEL 58	2006.2.02	-	-	B	-
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätskenivå sludge tank.

Table 25 Reliability Data Sensor Group Vättskenivå sludge tank

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	6.56E-08 <i>7.83E-08</i>	8.30E-08 <i>8.30E-08</i>	1.27E-08 <i>0.00E+00</i>	7.27E-08 <i>7.27E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		71.9
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

5.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-061 SIL2 Vättskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566) Safety Instrumented Function is quantified as follows.

Table 26 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.10E-08	3	N/A	72.47	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 27 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

5.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-061 SIL2 Vättskenivå sludge tank (345LS312|L2) => Close Avblödning slamvatten (345SV566) Safety Instrumented Function is quantified as follows.

Table 28 Final Element Part Functional Safety Performance

PFD AVG	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

5.3.3.1 Final Element Group 1: Close Avblödning slamvatten

The information and reliability data underneath describe the Close Avblödning slamvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcestor Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avblödning slamvatten.

Table 29 Reliability Data Final Element Group Close Avblödning slamvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

6 SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function.

6.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV634)
SIF Tag	SIF-062
SIF Description:	SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV634)

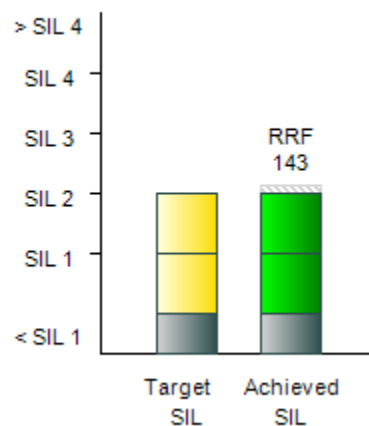
6.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 143.6



6.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

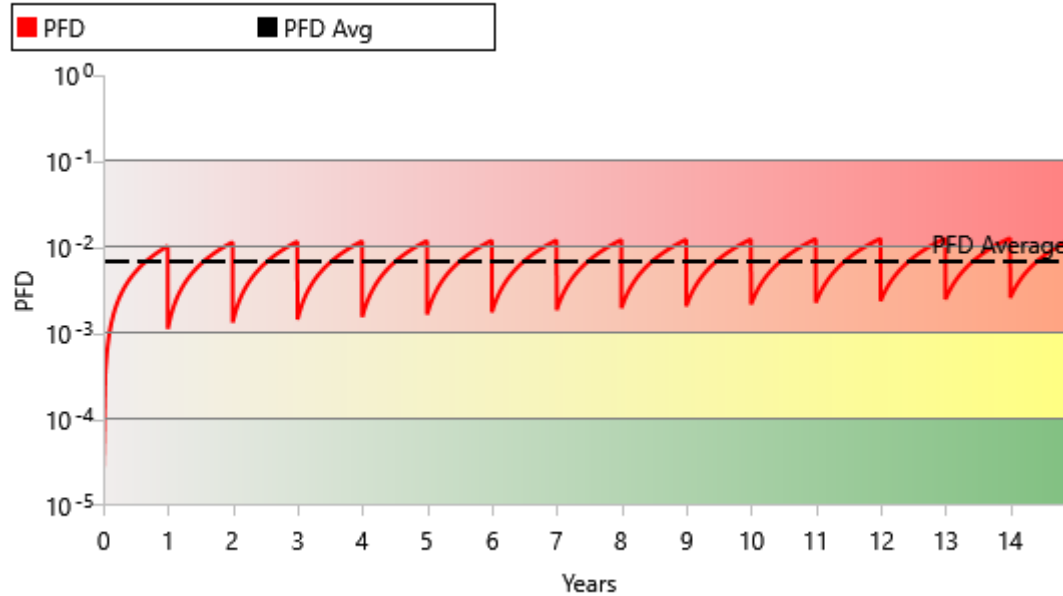
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

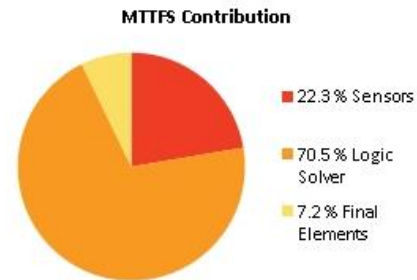
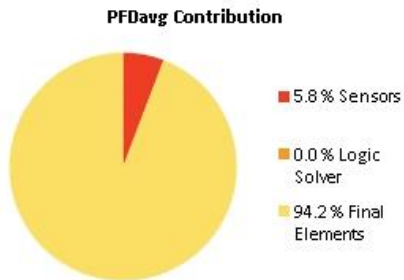
SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634)



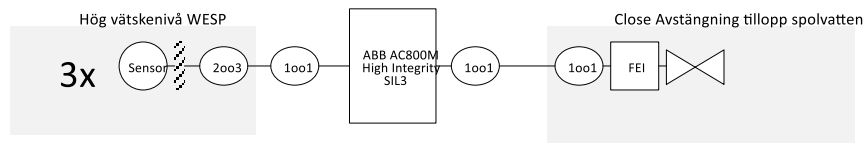
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 30 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
6.96E-03	143.6	2	2	N/A	51.06



The SIF Conceptual Design is modeled as shown in the picture below.



6.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function is quantified as follows.

Table 31 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.05E-04	3	N/A	228.67	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

6.3.1.1 Sensor Group 1: Hög vätskenivå WESP

The information and reliability data underneath describe the Hög vätskenivå WESP sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each): „ SERH VERSION 2H PIU AC TYPE SIL CAP

Enderess Hauser FMR50-57	-	-	B	3
Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off				
PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	B	3
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög vätskenivå WESP.

Table 32 Reliability Data Sensor Group Hög vätskenivå WESP

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Enderess Hauser FMR50-57	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	2.33E-06 <i>2.33E-06</i>	0.00E+00 <i>0.00E+00</i>	2.49E-07 <i>2.49E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>2.70E-08</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>1.87E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		84.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

6.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function is quantified as follows.

Table 33 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β -factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 34 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

6.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634) Safety Instrumented Function is quantified as follows.

Table 35 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

6.3.3.1 Final Element Group 1: Close Avstängning tillopp spolvatten

The information and reliability data underneath describe the Close Avstängning tillopp spolvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängning tillopp spolvatten.

Table 36 Reliability Data Final Element Group Close Avstängning tillopp spolvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

7 SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) Safety Instrumented Function.

7.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)
SIF Tag	SIF-063
SIF Description:	SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)

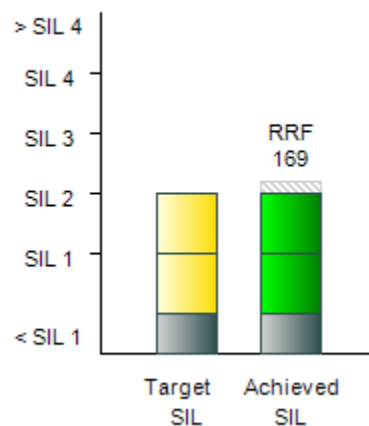
7.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 169.1



7.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

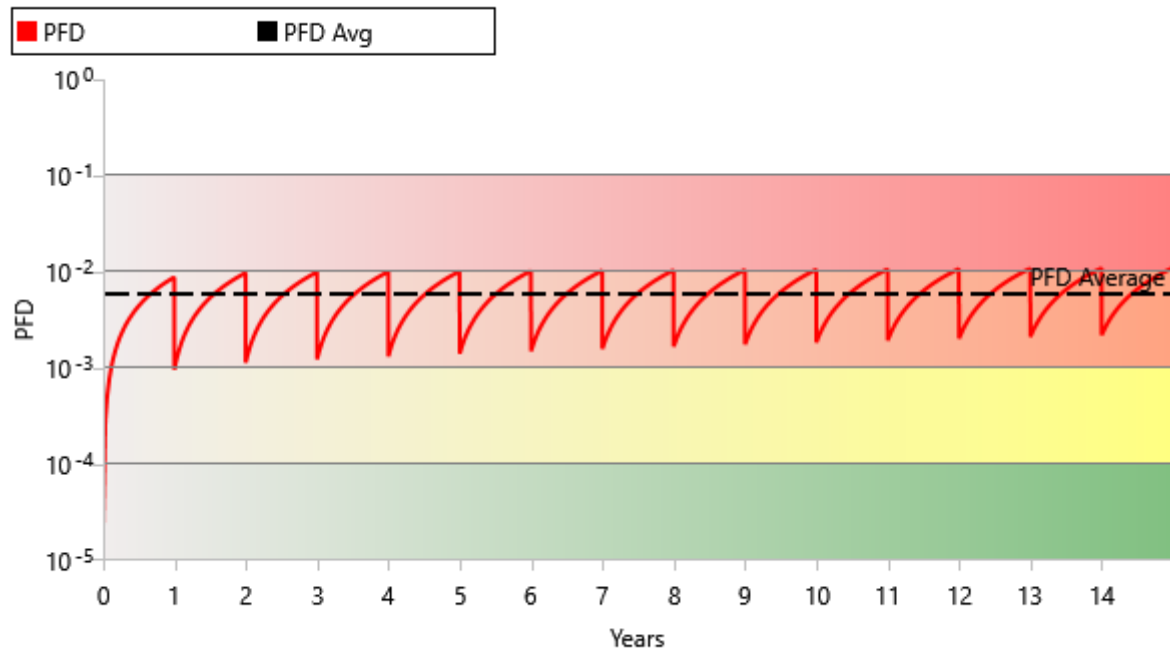
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)



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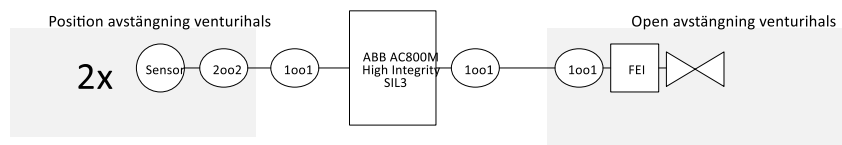
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 37 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
5.91E-03	169.1	2	2	N/A	60.42



The SIF Conceptual Design is modeled as shown in the picture below.



7.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) Safety Instrumented Function is quantified as follows.

Table 38 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.02E-04	2	N/A	40131.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

7.3.1.1 Sensor Group 1: Position avstängning venturihals

The information and reliability data underneath describe the Position avstängning venturihals sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg (each): , SERH VERSION 2H PIU AC TYPE SIL CAP

KControls 007

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [DigitalIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Position avstängning venturihals.

Table 39 Reliability Data Sensor Group Position avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
KControls 007	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.78E-08 <i>1.78E-08</i>	0.00E+00 <i>0.00E+00</i>	1.80E-08 <i>1.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>7.19E-09</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>1.66E-08</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
							SFF [%]		71.4	
							ROUTE 2H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

7.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) Safety Instrumented Function is quantified as follows.

Table 40 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.04E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT:	1	β -factor [%]:	1
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 41 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

7.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) Safety Instrumented Function is quantified as follows.

Table 42 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.71E-03	2	N/A	372.73	0	2

Number of Final Element group(s):	1
Voting between groups:	1oo1
β -factor [%]:	N/A

7.3.3.1 Final Element Group 1: Open avstängning venturihals

The information and reliability data underneath describe the Open avstängning venturihals final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β -factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open avstängning venturihals.

Table 43 Reliability Data Final Element Group Open avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599		1.46E-07		3.22E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		24.0
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

8 SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) Safety Instrumented Function.

8.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)
SIF Tag	SIF-064
SIF Description:	SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)

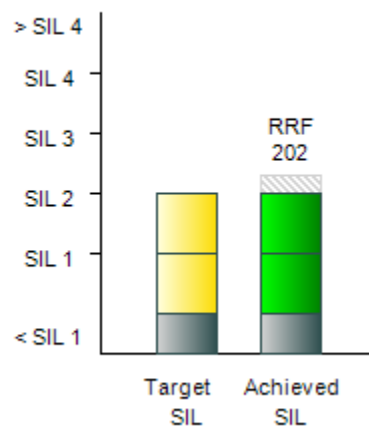
8.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 202.1



8.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

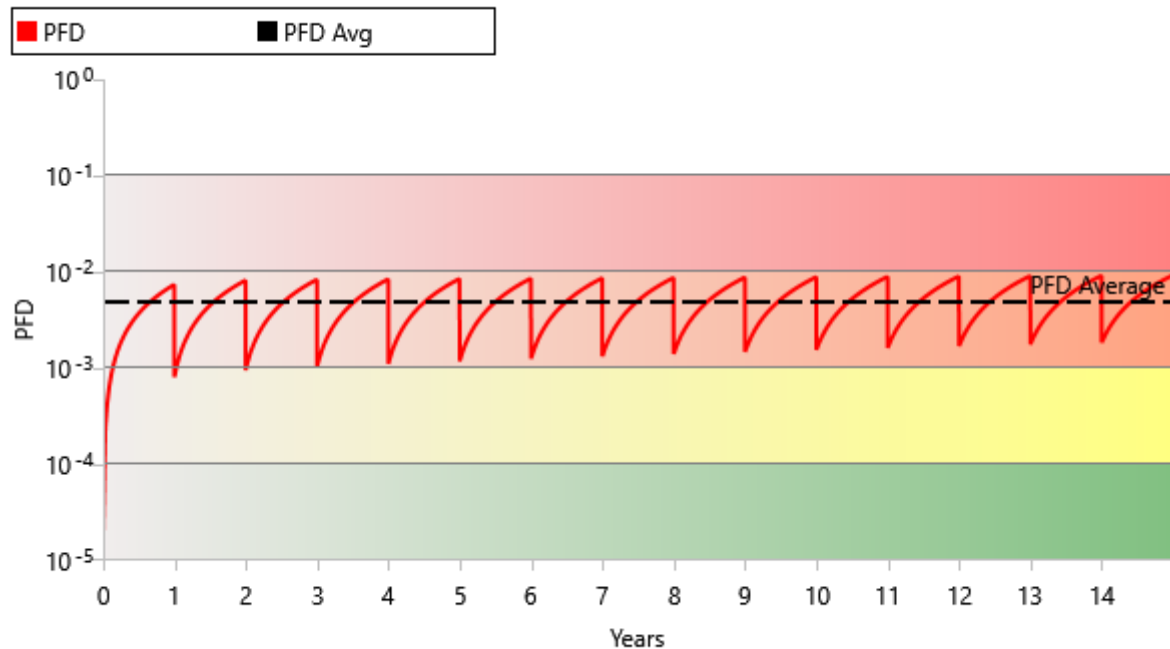
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)



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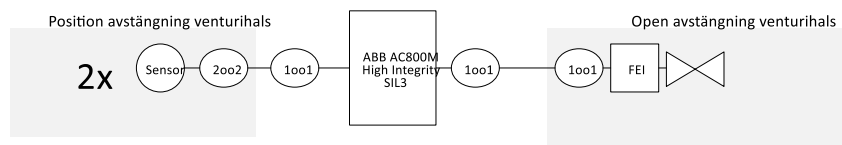
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 44 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.95E-03	202.1	2	2	N/A	65.45



The SIF Conceptual Design is modeled as shown in the picture below.



8.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) Safety Instrumented Function is quantified as follows.

Table 45 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.02E-04	2	N/A	40131.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

8.3.1.1 Sensor Group 1: Position avstängning venturihals

The information and reliability data underneath describe the Position avstängning venturihals sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg (each): , SERH VERSION 2H PIU AC TYPE SIL CAP

KControls 007

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [DigitalIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Position avstängning venturihals.

Table 46 Reliability Data Sensor Group Position avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
KControls 007	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.78E-08 <i>1.78E-08</i>	0.00E+00 <i>0.00E+00</i>	1.80E-08 <i>1.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>7.19E-09</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>1.66E-08</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		71.4
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

8.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) Safety Instrumented Function is quantified as follows.

Table 47 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.04E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β -factor [%]: 1
 MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
 Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 48 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

8.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) Safety Instrumented Function is quantified as follows.

Table 49 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.75E-03	2	N/A	707.89	0	2

Number of Final Element group(s): 1
 Voting between groups: 1oo1
 β -factor [%]: N/A

8.3.3.1 Final Element Group 1: Open avstängning venturihals

The information and reliability data underneath describe the Open avstängning venturihals final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
 HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N	-	-	-	A	3

Sirca AP-APM	-	-	A	3
GEFA HG1	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open avstängning venturihals.

Table 50 Reliability Data Final Element Group Open avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	16.7
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

9 SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) Safety Instrumented Function.

9.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)
SIF Tag	SIF-065
SIF Description:	SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)

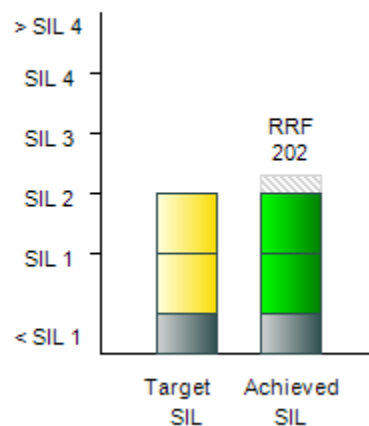
9.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 202.1



9.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

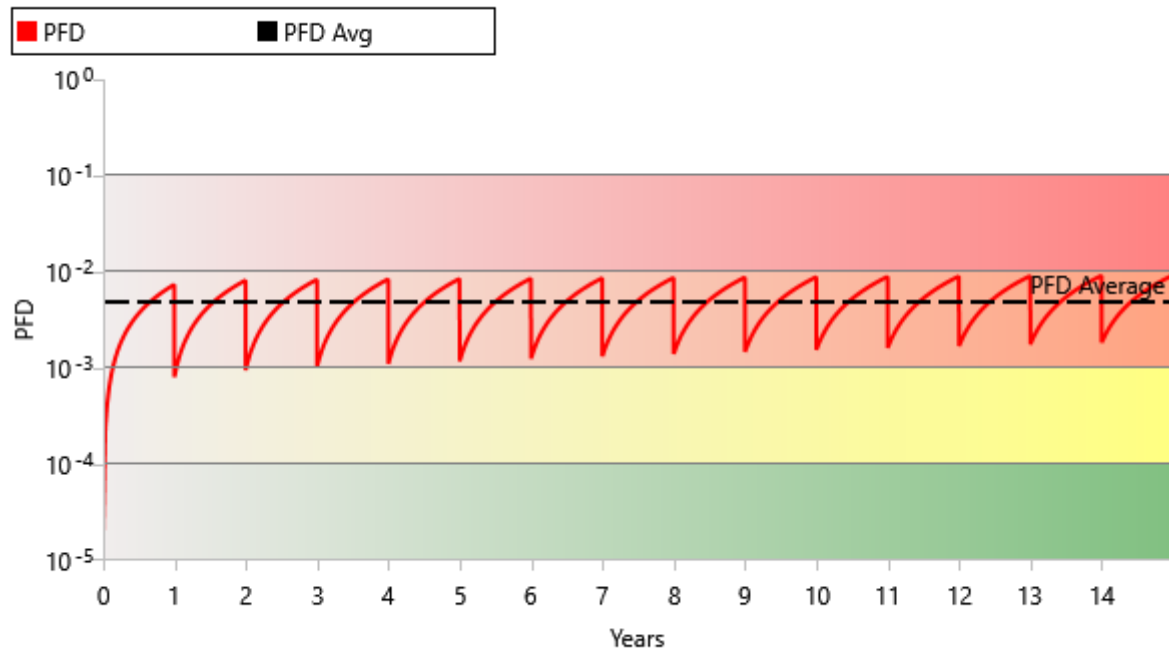
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)



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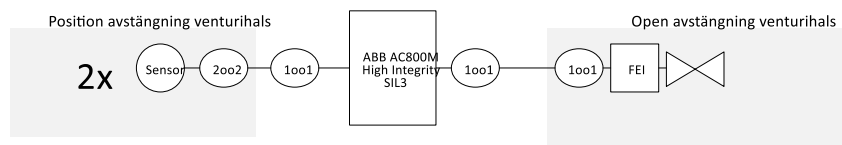
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 51 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.95E-03	202.1	2	2	N/A	65.45



The SIF Conceptual Design is modeled as shown in the picture below.



9.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) Safety Instrumented Function is quantified as follows.

Table 52 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.02E-04	2	N/A	40131.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

9.3.1.1 Sensor Group 1: Position avstängning venturihals

The information and reliability data underneath describe the Position avstängning venturihals sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg (each): , **SERH VERSION** **2H** **PIU** **AC TYPE** **SIL CAP**

KControls 007

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [DigitalIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Position avstängning venturihals.

Table 53 Reliability Data Sensor Group Position avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
KControls 007	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.78E-08 <i>1.78E-08</i>	0.00E+00 <i>0.00E+00</i>	1.80E-08 <i>1.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>7.19E-09</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>1.66E-08</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		71.4
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

9.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) Safety Instrumented Function is quantified as follows.

Table 54 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.04E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3

SERH VERSION 2013.1.02
PIU -
SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT:	1	β -factor [%]:	1
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 55 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

9.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) Safety Instrumented Function is quantified as follows.

Table 56 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.75E-03	2	N/A	707.89	0	2

Number of Final Element group(s):	1
Voting between groups:	1oo1
β -factor [%]:	N/A

9.3.3.1 Final Element Group 1: Open avstängning venturihals

The information and reliability data underneath describe the Open avstängning venturihals final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β -factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM	-	-	A	3
GEFA HG1	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open avstängning venturihals.

Table 57 Reliability Data Final Element Group Open avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	16.7
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

10 SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) =>

Close Avstängning tillopp spolvatten (345SV634)

Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function.

10.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)
SIF Tag	SIF-066
SIF Description:	SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)

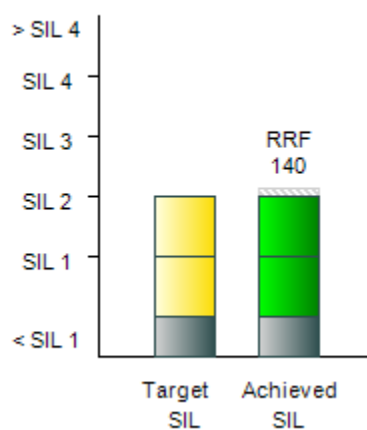
10.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 140.8



10.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) =>

Close Avstängning tillopp spolvatten (345SV634)

Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

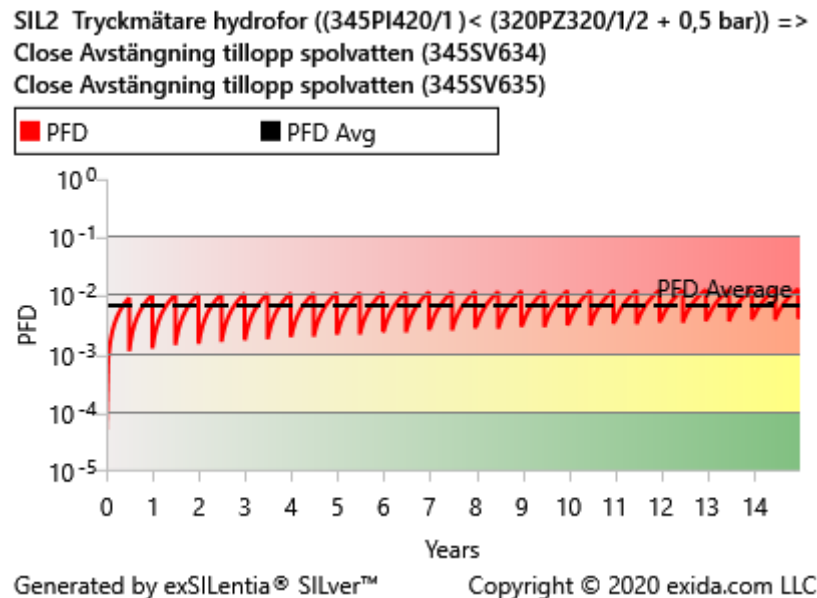
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:



Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) =>

Close Avstängning tillopp spolvatten (345SV634)

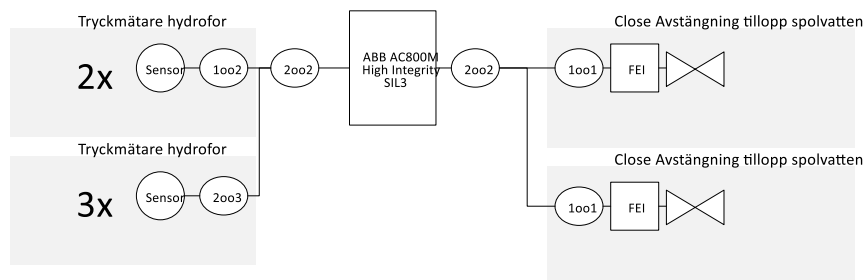
Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 58 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.10E-03	140.8	2	2	N/A	65.13



The SIF Conceptual Design is modeled as shown in the picture below.



10.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) =>

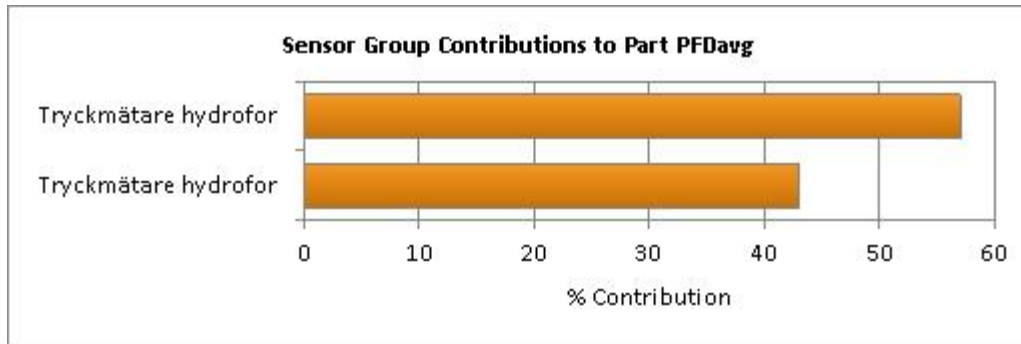
Close Avstängning tillopp spolvatten (345SV634)

Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 59 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.77E-04	3	N/A	1785.74	1	2

Number of Sensor group(s): 2
Voting between groups: 2oo2
β-factor [%]: 10



10.3.1.1 Sensor Group 1: Tryckmätare hydrofor

The information and reliability data underneath describe the Tryckmätare hydrofor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor: 10
[%]:

Equipment Leg
(each): ,

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckmätare hydrofor.

Table 60 Reliability Data Sensor Group Tryckmätare hydrofor

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

10.3.1.2 Sensor Group 2: Tryckmätare hydrofor

The information and reliability data underneath describe the Tryckmätare hydrofor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	2oo3	Voting type:	Identical					
HFT:	1	β-factor [%]:	10					
Equipment Leg (each):	„			SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75				2006.2.02	-	-	B	2
	Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off							
ABB AC800M High Integrity SIL3 [AnalogIn]				2013.1.02	-	-	B	3
MRT [Hours]:	24							
Proof Test Interval [Months]:		12		Proof Test Coverage [%]:			99	
Proof Test Execution:		Duration 1 hours						

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckmätare hydrofor.

Table 61 Reliability Data Sensor Group Tryckmätare hydrofor

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	78.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

10.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) =>

Close Avstängning tillopp spolvatten (345SV634)

Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 62 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.00E-08	3	N/A	72.45	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 63 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

10.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) =>

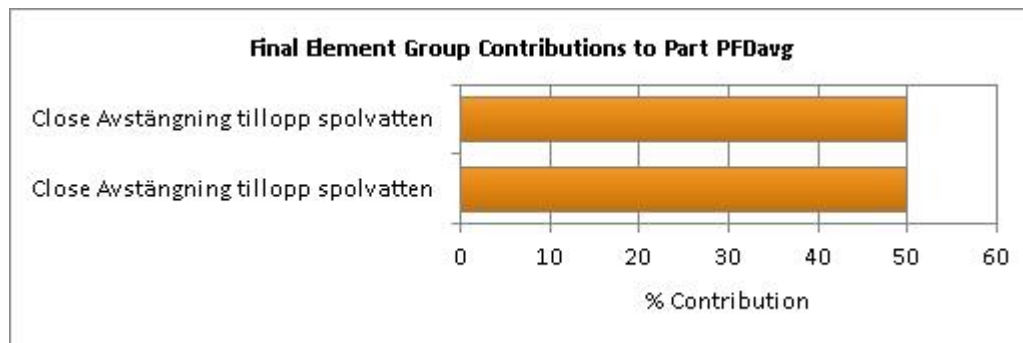
Close Avstängning tillopp spolvatten (345SV634)

Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 64 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.93E-03	2	N/A	1010.16	0	2

Number of Final Element group(s): 2
 Voting between groups: 2oo2
 β -factor [%]: 10



10.3.3.1 Final Element Group 1: Close Avstängning tillopp spolvatten

The information and reliability data underneath describe the Close Avstängning tillopp spolvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
 HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
 Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
 Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängning tillopp spolvatten.

Table 65 Reliability Data Final Element Group Close Avstängning tillopp spolvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

10.3.3.2 Final Element Group 2: Close Avstängning tillopp spolvatten

The information and reliability data underneath describe the Close Avstängning tillopp spolvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β-factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängning tillopp spolvatten.

Table 66 Reliability Data Final Element Group Close Avstängning tillopp spolvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

11 SIF-067 SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högsäpänningsenhet (353EC015)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-067 SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högsäpänningsenhet (353EC015) Safety Instrumented Function.

11.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högsäpänningsenhet (353EC015)
SIF Tag	SIF-067
SIF Description:	SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högsäpänningsenhet (353EC015)

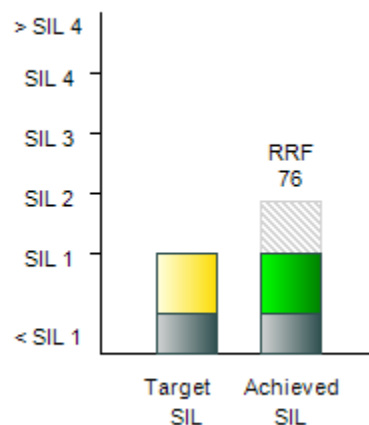
11.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 76.7



11.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-067 SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högsäpänningsenhet (353EC015) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

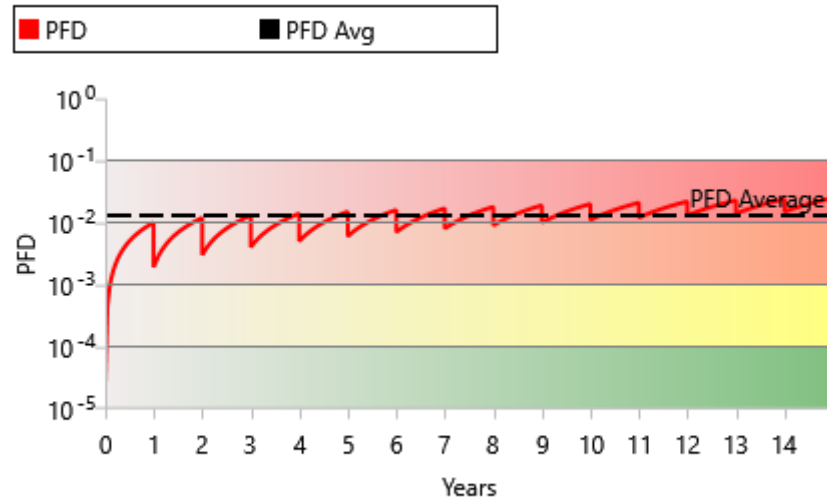
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högsäpänningsenhet (353EC015)



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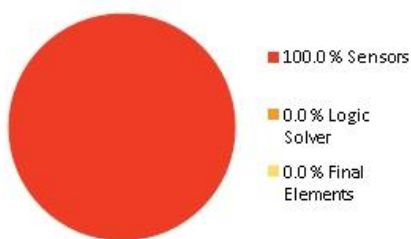
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-067 SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högsäpänningsenhet (353EC015) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

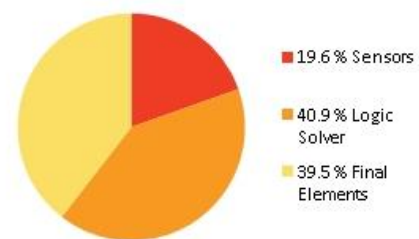
Table 67 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.30E-02	76.7	1	2	N/A	29.66

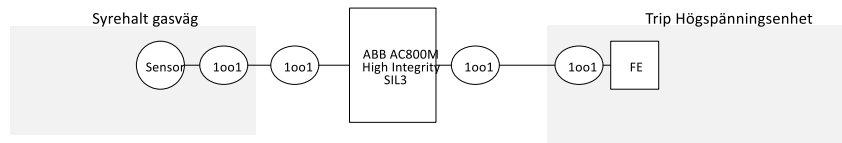
PFDavg Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



11.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-067 SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högspänningsenhet (353EC015) Safety Instrumented Function is quantified as follows.

Table 68 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.30E-02	2	N/A	151.42	0	2

Number of Sensor group(s): 1
 Voting between groups: 1001
 β-factor [%]: N/A

11.3.1.1 Sensor Group 1: Syrehalt gasväg

The information and reliability data underneath describe the Syrehalt gasväg sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
 HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Siemens Sitrans SL		-	-	B	1
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 72					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Syrehalt gasväg.

Table 69 Reliability Data Sensor Group Syrehalt gasvåg

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Siemens Sitrans SL	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.41E-07 <i>7.41E-07</i>	0.00E+00 <i>0.00E+00</i>	1.16E-06 <i>1.16E-06</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		39.6
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

11.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-067 SIL1 Syrehalt gasvåg (353AZ362|H2) => Trip Högsäpningssenhett (353EC015) Safety Instrumented Function is quantified as follows.

Table 70 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.10E-08	3	N/A	72.47	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 71 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

11.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-067 SIL1 Syrehalt gasväg (353AZ362|H2) => Trip Högsäpänningsenhet (353EC015) Safety Instrumented Function is quantified as follows.

Table 72 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.15E-06	2	N/A	75.1	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

11.3.3.1 Final Element Group 1: Trip Högsäpänningsenhet

The information and reliability data underneath describe the Trip Högsäpänningsenhet final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
PSR-PS20		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip Högsäpänningsenhet.

Table 73 Reliability Data Final Element Group Trip Högspänningsenhet

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
PSR-PS20		1.42E-06		1.95E-10			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

12 SIF-068 SIL2 Vätsketryck nedströms pump ($353PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(353SV566)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-068 SIL2 Vätsketryck nedströms pump ($353PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(353SV566) Safety Instrumented Function.

12.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Vätsketryck nedströms pump ($353PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(353SV566)
SIF Tag	SIF-068
SIF Description:	SIL2 Vätsketryck nedströms pump ($353PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(353SV566)

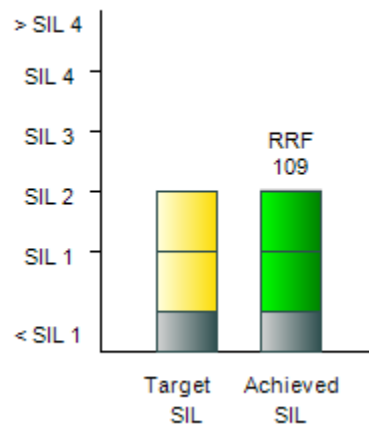
12.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 109.3



12.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-068 SIL2 Vätsketryck nedströms pump ($353PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(353SV566) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

exSILentia-VER-002-01-b SILver Detailed-Eng

Mission Time: 10 years
Startup time: 24 hours

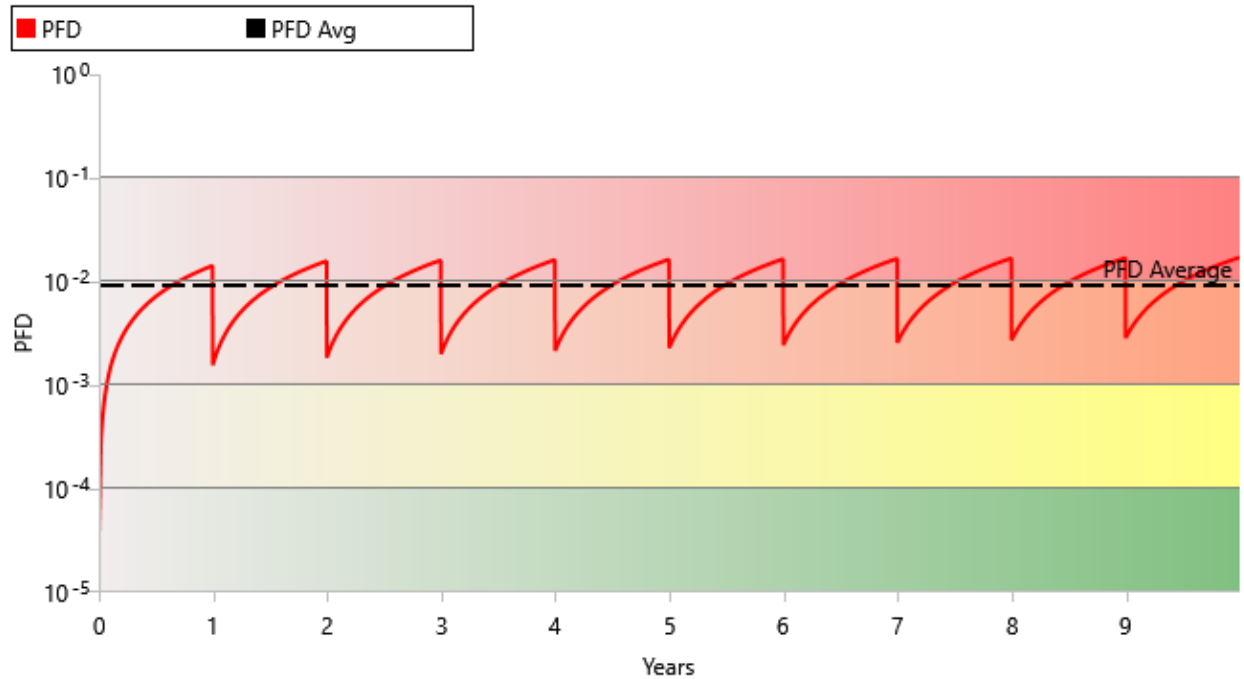
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566)



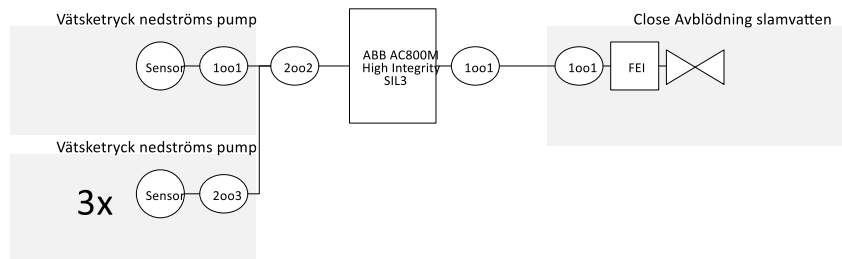
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-068 SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 74 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
9.15E-03	109.3	2	2	N/A	65.33



The SIF Conceptual Design is modeled as shown in the picture below.



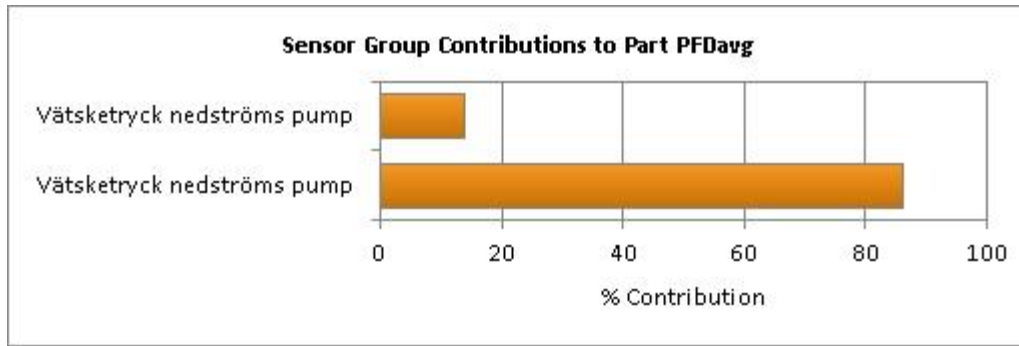
12.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-068 SIL2 Vätsketryck nedströms pump ($353PI326 < (320PZ320/1/2 + 0,5 \text{ bar})$) => Close Avblödning slamvatten(353SV566) Safety Instrumented Function is quantified as follows.

Table 75 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.89E-03	2	N/A	10494.91	0	2

Number of Sensor group(s): 2
Voting between groups: 2oo2
β-factor [%]:



12.3.1.1 Sensor Group 1: Vätsketryck nedströms pump

The information and reliability data underneath describe the Vätsketryck nedströms pump sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätsketryck nedströms pump.

Table 76 Reliability Data Sensor Group Vätsketryck nedströms pump

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	18.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

12.3.1.2 Sensor Group 2: Vätsketryck nedströms pump

The information and reliability data underneath describe the Vätsketryck nedströms pump sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each): „

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätsketryck nedströms pump.

Table 77 Reliability Data Sensor Group Vätsketryck nedströms pump

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	18.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

12.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-068 SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566) Safety Instrumented Function is quantified as follows.

Table 78 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.92E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 79 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

12.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-068 SIL2 Vätstektryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566) Safety Instrumented Function is quantified as follows.

Table 80 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.28E-03	2	N/A	708.98	0	2

Number of Final Element group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

12.3.3.1 Final Element Group 1: Close Avblödning slamvatten

The information and reliability data underneath describe the Close Avblödning slamvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worchester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avblödning slamvatten.

Table 81 Reliability Data Final Element Group Close Avblödning slamvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worchester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	12.6
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

13 SIF-069 SIL2 Vätskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-069 SIL2 Vätskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function.

13.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635)
SIF Tag	SIF-069
SIF Description:	SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635)

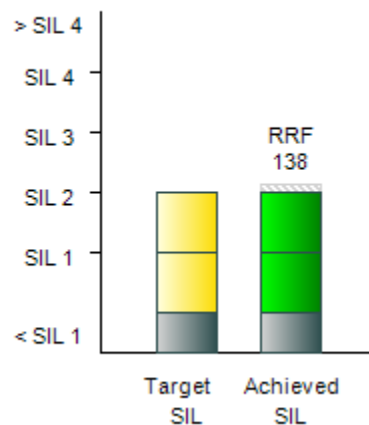
13.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 138.4



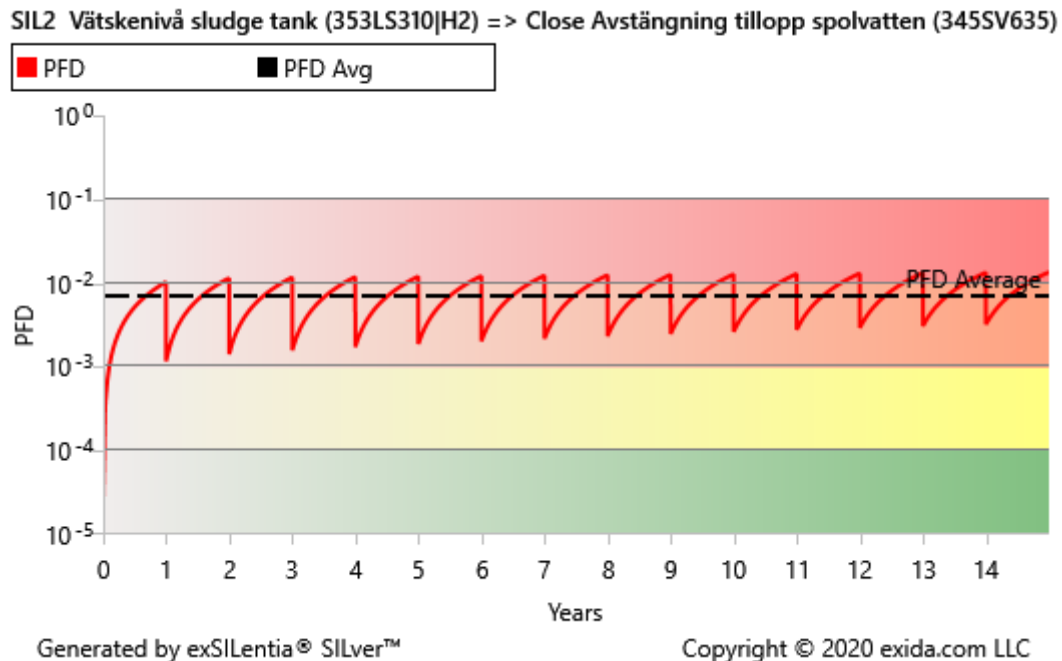
13.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-069 SIL2 Vätskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

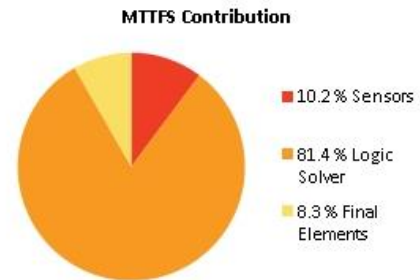
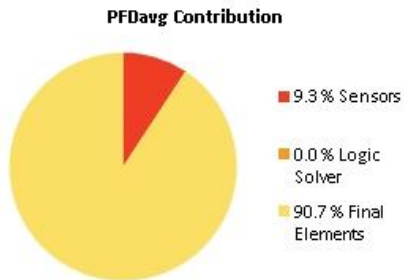
Analyst: Analysis Date: 10/4/2019
Design References
Comments:



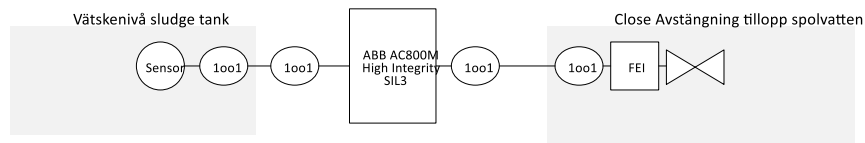
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-069 SIL2 Vätskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 82 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.22E-03	138.4	2	2	N/A	59.02



The SIF Conceptual Design is modeled as shown in the picture below.



13.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-069 SIL2 Vätskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 83 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.69E-04	2	N/A	576.96	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

13.3.1.1 Sensor Group 1: Vätskenivå sludge tank

The information and reliability data underneath describe the Vätskenivå sludge tank sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

Endress+Hauser Liquiphant M/S + FEL 58 2006.2.02 - - B -

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vättskenivå sludge tank.

Table 84 Reliability Data Sensor Group Vättskenivå sludge tank

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	8.39E-08 <i>9.01E-08</i>	8.53E-08 <i>8.53E-08</i>	6.21E-09 <i>0.00E+00</i>	5.92E-08 <i>5.92E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	77.2
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

13.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-069 SIL2 Vättskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 85 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.10E-08	3	N/A	72.47	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β -factor [%]: 1
 MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
 Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 86 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

13.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-069 SIL2 Vättskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 87 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
 Voting between groups: 1oo1
 β -factor [%]: N/A

13.3.3.1 Final Element Group 1: Close Avstängning tillopp spolvatten

The information and reliability data underneath describe the Close Avstängning tillopp spolvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
 HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängning tillopp spolvatten.

Table 88 Reliability Data Final Element Group Close Avstängning tillopp spolvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

14 SIF-070 SIL2 Vätskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-070 SIL2 Vätskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566) Safety Instrumented Function.

14.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566)
SIF Tag	SIF-070
SIF Description:	SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566)

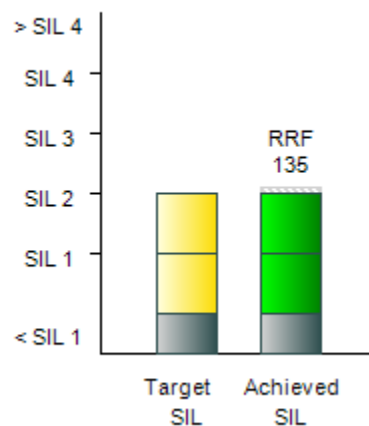
14.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 135.5



14.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-070 SIL2 Vätskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

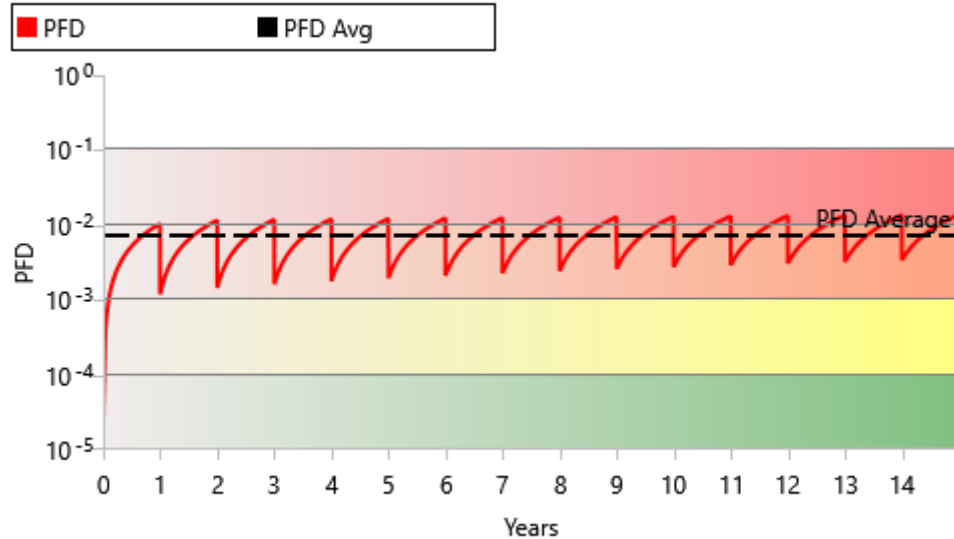
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Vätskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566)

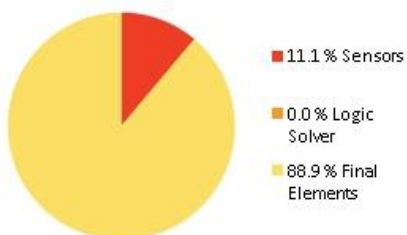


Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-070 SIL2 Vätskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

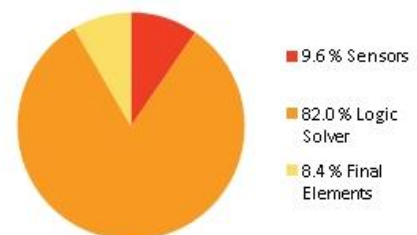
Table 89 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.38E-03	135.5	2	2	N/A	59.46

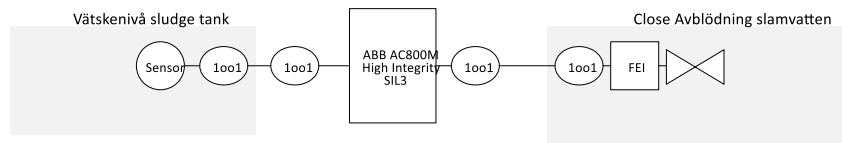
PFDavg Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



14.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-070 SIL2 Vätskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566) Safety Instrumented Function is quantified as follows.

Table 90 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.22E-04	2	N/A	621.33	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

14.3.1.1 Sensor Group 1: Vätskenivå sludge tank

The information and reliability data underneath describe the Vätskenivå sludge tank sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Liquiphant M/S + FEL 58	2006.2.02	-	-	B	-
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätskenivå sludge tank.

Table 91 Reliability Data Sensor Group Vättskenivå sludge tank

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	6.56E-08 <i>7.83E-08</i>	8.30E-08 <i>8.30E-08</i>	1.27E-08 <i>0.00E+00</i>	7.27E-08 <i>7.27E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		71.9
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

14.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-070 SIL2 Vättskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566) Safety Instrumented Function is quantified as follows.

Table 92 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.10E-08	3	N/A	72.47	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 93 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

14.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-070 SIL2 Vättskenivå sludge tank (353LS312|L2) => Close Avblödning slamvatten (353SV566) Safety Instrumented Function is quantified as follows.

Table 94 Final Element Part Functional Safety Performance

PFD AVG	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

14.3.3.1 Final Element Group 1: Close Avblödning slamvatten

The information and reliability data underneath describe the Close Avblödning slamvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcestor Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avblödning slamvatten.

Table 95 Reliability Data Final Element Group Close Avblödning slamvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

15 SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function.

15.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635)
SIF Tag	SIF-071
SIF Description:	SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635)

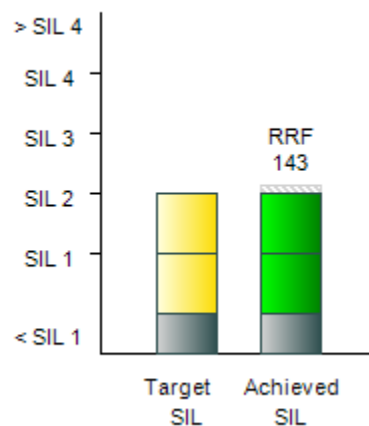
15.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 143.6



15.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

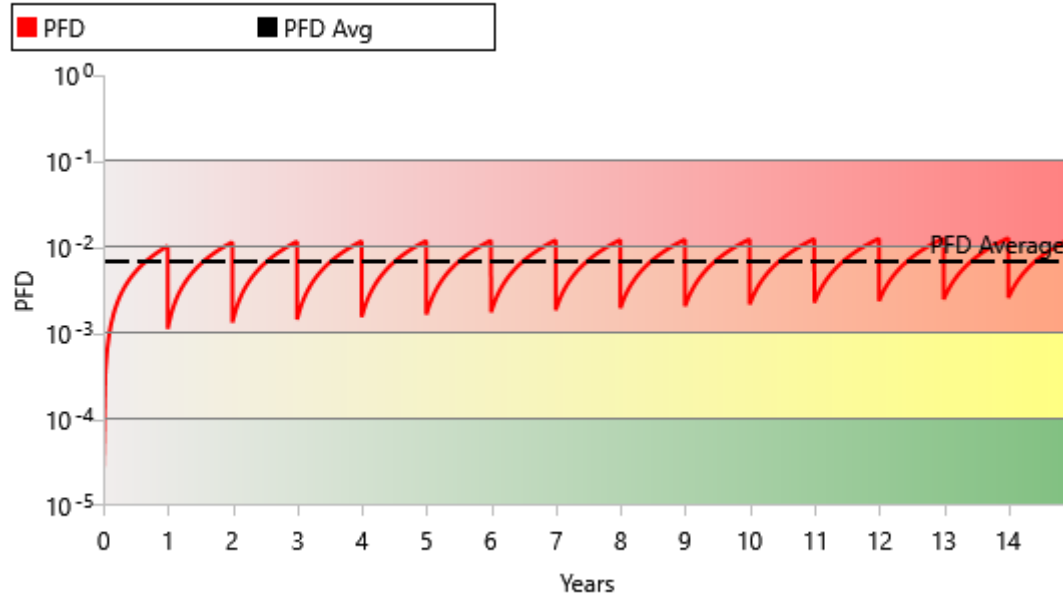
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

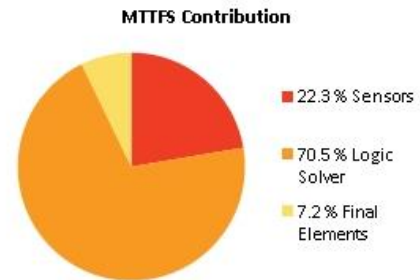
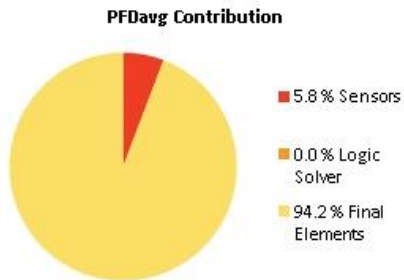
SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635)



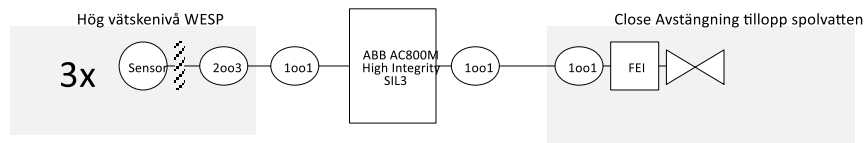
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 96 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
6.96E-03	143.6	2	2	N/A	51.06



The SIF Conceptual Design is modeled as shown in the picture below.



15.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 97 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.05E-04	3	N/A	228.67	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

15.3.1.1 Sensor Group 1: Hög vätskenivå WESP

The information and reliability data underneath describe the Hög vätskenivå WESP sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each): „ SERH VERSION 2H PIU AC TYPE SIL CAP

Enderess Hauser FMR50-57	-	-	B	3
Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off				
PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	B	3
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög vätskenivå WESP.

Table 98 Reliability Data Sensor Group Hög vätskenivå WESP

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Enderess Hauser FMR50-57	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	2.33E-06 <i>2.33E-06</i>	0.00E+00 <i>0.00E+00</i>	2.49E-07 <i>2.49E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>2.70E-08</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>1.87E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		84.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

15.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 99 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B		
HFT:	β -factor [%]:	1		
MRT [Hours]:				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 100 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

15.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635) Safety Instrumented Function is quantified as follows.

Table 101 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.56E-03	2	N/A	709.18	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

15.3.3.1 Final Element Group 1: Close Avstängning tillopp spolvatten

The information and reliability data underneath describe the Close Avstängning tillopp spolvatten final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängning tillopp spolvatten.

Table 102 Reliability Data Final Element Group Close Avstängning tillopp spolvatten

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599				4.68E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		12.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

16 SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) Safety Instrumented Function.

16.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)
SIF Tag	SIF-072
SIF Description:	SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)

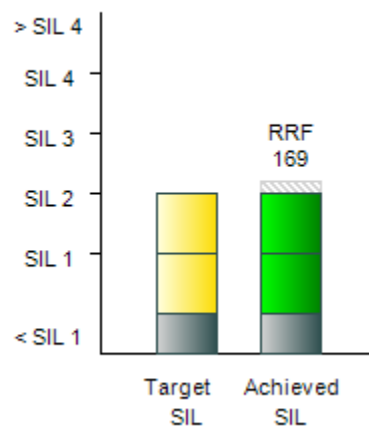
16.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 169.1



16.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

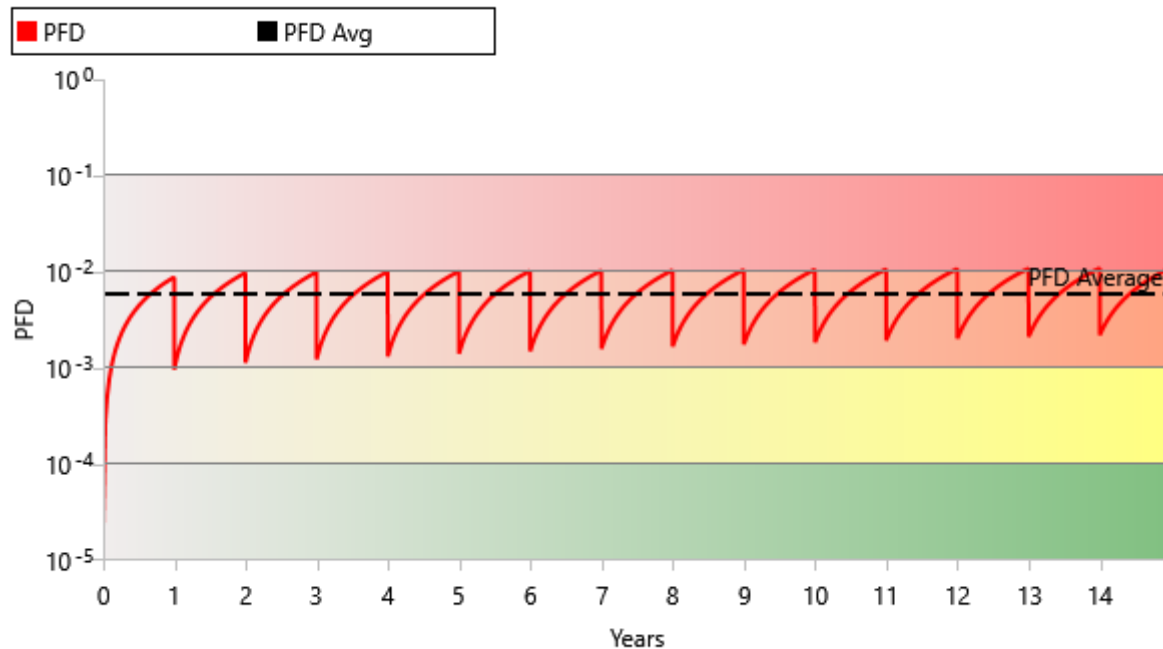
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)



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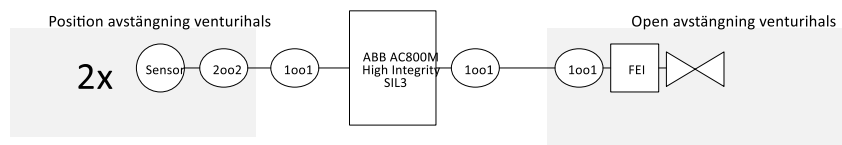
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 103 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
5.91E-03	169.1	2	2	N/A	60.42



The SIF Conceptual Design is modeled as shown in the picture below.



16.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) Safety Instrumented Function is quantified as follows.

Table 104 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.02E-04	2	N/A	40131.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

16.3.1.1 Sensor Group 1: Position avstängning venturihals

The information and reliability data underneath describe the Position avstängning venturihals sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg (each): , **SERH VERSION** **2H** **PIU** **AC TYPE** **SIL CAP**

KControls 007

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [DigitalIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Position avstängning venturihals.

Table 105 Reliability Data Sensor Group Position avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
KControls 007	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.78E-08 <i>1.78E-08</i>	0.00E+00 <i>0.00E+00</i>	1.80E-08 <i>1.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>7.19E-09</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>1.66E-08</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		71.4
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

16.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) Safety Instrumented Function is quantified as follows.

Table 106 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.04E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β -factor [%]: 1
MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 107 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

16.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) Safety Instrumented Function is quantified as follows.

Table 108 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.71E-03	2	N/A	372.73	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

16.3.3.1 Final Element Group 1: Open avstängning venturihals

The information and reliability data underneath describe the Open avstängning venturihals final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open avstängning venturihals.

Table 109 Reliability Data Final Element Group Open avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599		1.46E-07		3.22E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	24.0
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

17 SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) Safety Instrumented Function.

17.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)
SIF Tag	SIF-073
SIF Description:	SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)

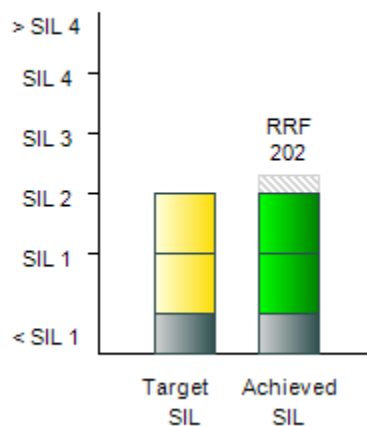
17.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 202.1



17.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

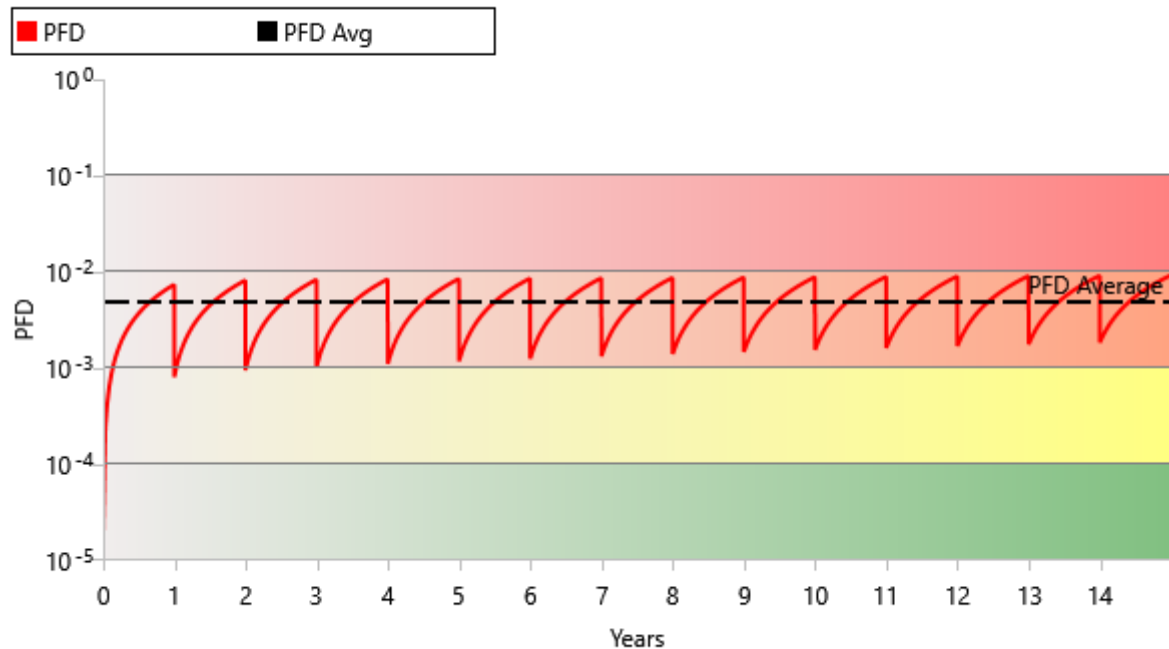
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)



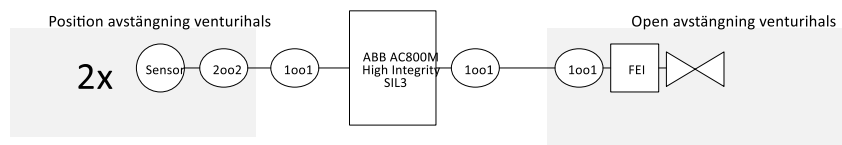
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 110 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.95E-03	202.1	2	2	N/A	65.45



The SIF Conceptual Design is modeled as shown in the picture below.



17.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) Safety Instrumented Function is quantified as follows.

Table 111 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.02E-04	2	N/A	40131.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

17.3.1.1 Sensor Group 1: Position avstängning venturihals

The information and reliability data underneath describe the Position avstängning venturihals sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg (each): , SERH VERSION 2H PIU AC TYPE SIL CAP

KControls 007

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [DigitalIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Position avstängning venturihals.

Table 112 Reliability Data Sensor Group Position avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
KControls 007	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.78E-08 <i>1.78E-08</i>	0.00E+00 <i>0.00E+00</i>	1.80E-08 <i>1.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>7.19E-09</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>1.66E-08</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		71.4
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

17.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) Safety Instrumented Function is quantified as follows.

Table 113 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.04E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT:	1	β -factor [%]:	1
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 114 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

17.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) Safety Instrumented Function is quantified as follows.

Table 115 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.75E-03	2	N/A	707.89	0	2

Number of Final Element group(s):	1
Voting between groups:	1oo1
β -factor [%]:	N/A

17.3.3.1 Final Element Group 1: Open avstängning venturihals

The information and reliability data underneath describe the Open avstängning venturihals final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β -factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM	-	-	A	3
GEFA HG1	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open avstängning venturihals.

Table 116 Reliability Data Final Element Group Open avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	16.7
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

18 SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) Safety Instrumented Function.

18.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)
SIF Tag	SIF-074
SIF Description:	SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)

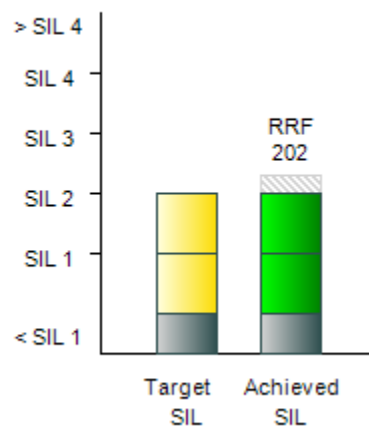
18.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 202.1



18.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

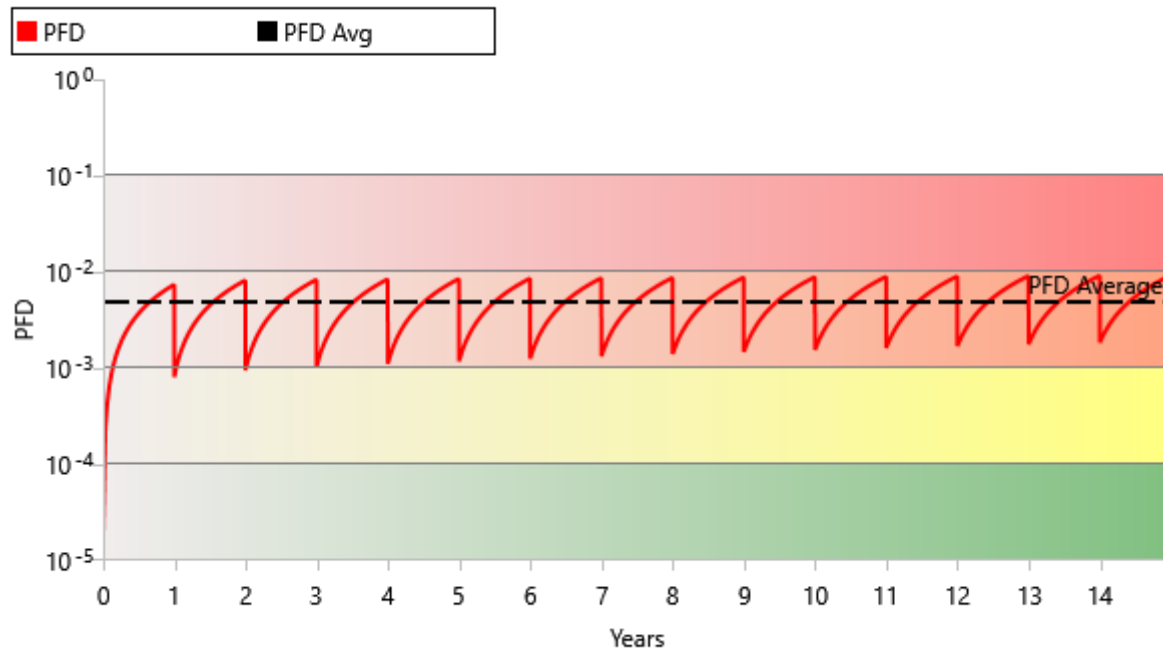
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)



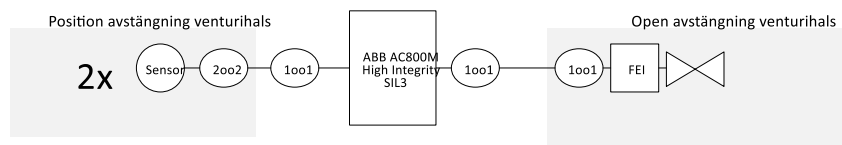
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 117 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.95E-03	202.1	2	2	N/A	65.45



The SIF Conceptual Design is modeled as shown in the picture below.



18.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) Safety Instrumented Function is quantified as follows.

Table 118 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.02E-04	2	N/A	40131.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

18.3.1.1 Sensor Group 1: Position avstängning venturihals

The information and reliability data underneath describe the Position avstängning venturihals sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg (each): , SERH VERSION 2_H PIU AC TYPE SIL CAP

KControls 007

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [DigitalIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Position avstängning venturihals.

Table 119 Reliability Data Sensor Group Position avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
KControls 007	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.78E-08 <i>1.78E-08</i>	0.00E+00 <i>0.00E+00</i>	1.80E-08 <i>1.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>7.19E-09</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>1.66E-08</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		71.4
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

18.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) Safety Instrumented Function is quantified as follows.

Table 120 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.04E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT:	1	β -factor [%]:	1
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 121 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

18.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) Safety Instrumented Function is quantified as follows.

Table 122 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.75E-03	2	N/A	707.89	0	2

Number of Final Element group(s):	1
Voting between groups:	1oo1
β -factor [%]:	N/A

18.3.3.1 Final Element Group 1: Open avstängning venturihals

The information and reliability data underneath describe the Open avstängning venturihals final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β -factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3

Sirca AP-APM	-	-	A	3
GEFA HG1	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open avstängning venturihals.

Table 123 Reliability Data Final Element Group Open avstängning venturihals

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	16.7
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

19 SILver™ Group Reuse Overview

This chapter provides an overview of the various groups that were defined during the SIL verification analyses of the Hybrit Pilot Plant project. The reuse of groups is available between Safety Instrumented Functions to account for identical hardware used in those different Safety Instrumented Functions. The group reuse allows users to specify a specific group once, reuse it, and with all future changes made to a single group cascading to all Safety Instrumented Functions it is used in. It is instrumental for the user to review all SIFs a single group is used in when changes are made to that group.

This overview indicates the groups that were reused, the number of times they were reused, and the Safety Instrumented Functions that they were used in.

19.1 Sensor Groups Reused

Gastrück toppgas (used 4 times)	SIF-013, SIF-014, SIF-011, SIF-012
Gastrück transferline (used 3 times)	SIF-021, SIF-024, SIF-020
Tryckvakt O2-inj (used 3 times)	SIF-022, SIF-023, SIF-020
Temperatur konvektionsdel (used 2 times)	SIF-027, SIF-028

19.2 Logic Solver Groups Reused

ABB AC800M High Integrity SIL3 (used 72 times)	SIF-996, SIF-997, SIF-998, SIF-026, SIF-001, SIF-034, SIF-002, SIF-003, SIF-004, SIF-005, SIF-051, SIF-052, SIF-053, SIF-054, SIF-056, SIF-058, SIF-059, SIF-060, SIF-061, SIF-062, SIF-063, SIF-076, SIF-077, SIF-064, SIF-065, SIF-067, SIF-068, SIF-069, SIF-070, SIF-071, SIF-074, SIF-073, SIF-072, SIF-075, SIF-019, SIF-013, SIF-014, SIF-017, SIF-018, SIF-081, SIF-039, SIF-040, SIF-042, SIF-045, SIF-046, SIF-048, SIF-011, SIF-021, SIF-024, SIF-022, SIF-023, SIF-020, SIF-025, SIF-027, SIF-028, SIF-029, SIF-015, SIF-012, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-037, SIF-038, SIF-041, SIF-043, SIF-044, SIF-047, SIF-080, SIF-079, SIF-082
ABB AC800M High Integrity SIL3 (used 9 times)	SIF-006, SIF-007, SIF-008, SIF-009, SIF-049, SIF-050, SIF-055, SIF-057, SIF-066

19.3 Final Element Groups Reused

Avstängning O2 (used 6 times)	SIF-026, SIF-037, SIF-043, SIF-044, SIF-079, SIF-082
Trip red.gaskompressor (used 6 times)	SIF-050, SIF-025, SIF-043, SIF-044, SIF-047, SIF-038
Trip proc.gaskompressor (used 5 times)	SIF-055, SIF-042, SIF-045, SIF-046, SIF-048
Trip red.gaskompressor (used 2 times)	SIF-049, SIF-041
Avstängningsventil 2 överhettad ånga till PGH (used 8 times)	SIF-076, SIF-016, SIF-017, SIF-018, SIF-028, SIF-041, SIF-047, SIF-080
H2 till brännare PGH (used 2 times)	SIF-019, SIF-079
NG till brännare PGH (used 2 times)	SIF-019, SIF-082
Close NG till kylkrets	SIF-042, SIF-045, SIF-046, SIF-048

(used 4 times)

Avstängning O2 PT12

(used 18 times)

NG till brännare PGH PT12

(used 5 times)

Avstängningsventil 2 överhettad ånga
till PGH PT 6

(used 5 times)

SIF-021, SIF-022, SIF-024, SIF-023, SIF-020, SIF-025, SIF-027, SIF-029,
SIF-015, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-041, SIF-047,
SIF-081, SIF-038

SIF-025, SIF-027, SIF-029, SIF-031, SIF-032

SIF-012, SIF-043, SIF-044, SIF-079, SIF-082

20 Abbreviations and Definitions

20.1 Abbreviations

HFT	Hardware Fault Tolerance
IPL	Independent Protection Layers
MoonN	"M" out of "N"
MTTFS	Mean Time To Fail Spurious
MRT	Mean Repair Time
PFD	Probability of Failure on Demand
PFDavg	Average Probability of Failure on Demand
PFH	Probability of a Dangerous Failure per Hour
PIU	Proven In Use
PTC	Proof Test Coverage
PTI	Proof Test Interval
RRF	Risk Reduction Factor
SERH	Safety Equipment Reliability Handbook
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SSI	Site Safety Index
SRS	Safety Requirements Specification
β -factor	Beta factor, indicating common cause susceptibility
SD	Safe Detected
SU	Safe Undetected
DD	Dangerous Detected
DU	Dangerous Undetected
AD	Annunciation Detected
AU	Annunciation Undetected
NE	No Effect

20.2 Definitions

Architectural Constraints	Indication of maximum SIL achievable of a SIF's subsystems architecture based on minimum hardware fault tolerance requirements. This is determined solely on the basis of Type A or Type B device selection, redundancy (hardware fault tolerance), and the safe failure fraction (calculated or conservatively assumed if no data is provided). It does not pertain to Systematic Capability or certification levels. Architectural Constraint limits are determined based on IEC 61508 or IEC 61511 tables.
Mission Time	This is the time period that the SIF is expected to be operational. Alternatively this is the time period between points in time where the SIF can be considered as new. Note that this is not the proof test interval because of imperfect proof tests considerations. The mission time could, for example, correspond to the major turnaround period of a unit.
Startup Time	This represents the time it will take to perform maintenance and repair on a unit after a spurious trip plus the time it will take to restart the unit. This is typically longer than just the average repair time.

Systematic Capability

Indication of the maximum SIL achievable based on the SIF's equipment items systematic integrity alone. This is determined on the basis of the IEC 61508 certification level of each equipment item and/or the systematic integrity claim made by the user for a specific equipment item as part of its Proven In use justification. This addresses the requirement in IEC 61511 that all equipment used in a SIF must either be assessed per IEC 61508 or proven in use.

21 Disclaimer, Assumptions, Equipment Data

21.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

21.2 Assumptions SILver™

SIL Capability

If the user disables the **Consider IEC SIL Capability** option, it is assumed that the justification of equipment used is performed outside of the SILver tool as in that case the results of the SIL verification only address the quantitative requirements of IEC 61511.

A list of all other assumptions on which the SILver™ tool is based can be found in the exSILentia® user guide.

21.3 Equipment Data

exida has compiled a proprietary equipment failure database. This database is a compilation of failure data collected from a variety of public and confidential sources and presents an industry average. The database is published as the "Safety Equipment Reliability Handbook" ISBN-13: 978-1-934977-15-6. The reliability data collection process as described in this book applies to the SILver equipment data collection process.

The user is responsible for determining the applicability of the failure data to any particular environment. The stress levels assumed to determine the equipment failure rate are average worst case for an industrial environment. Descriptions of the environmental profiles used are included in the Safety Equipment Reliability Handbook. Accurate plant specific data is preferable to general industry average data. Industrial plant sites with high levels of stress must use failure rate data that is adjusted to a higher value to account for the specific conditions of the plant.